

ZIHAO WANG

1 Tsinghuayuan, Haidian District, Beijing, China
+86 184 3888 2780 • w-zihao24@mails.tsinghua.edu.cn

EDUCATION

- M.S. in Mechanical Engineering, Tsinghua University** *Sep. 2024–Present*
Department of Mechanical Engineering, Beijing, China
• GPA: 3.76/4.00
• Relevant Coursework: Digital Image Processing and Applications; Deep Learning; Intelligent Mechatronic Control Engineering
- B.Eng. in Mechanical Engineering, Tsinghua University** *Sep. 2020–Jun. 2024*
Department of Mechanical Engineering, Beijing, China
• GPA: 3.57/4.00; ranked 12/75
• Honors: Academic Excellence Award, Tsinghua University; National Encouragement Scholarship (top 10%)
• Relevant Coursework: Fundamentals of Computer Programming; Practice of Mechatronic Control Systems
- Minor Coursework in Software Engineering, Tsinghua University** *Sep. 2021–Jun. 2023*
• Object-Oriented Programming; Discrete Mathematics I and II; Fundamentals of Programming

RESEARCH EXPERIENCE

- Vision-Based Intelligent Pipeline Welding System** *Sep. 2023–Present*
Beijing Municipal Science and Technology Commission Project, Beijing, China
• Developed robotic welding control models and optimized key parameters for automated pipeline welding.
• Constructed 20-frame temporal samples from large-scale sensor datasets and developed an end-to-end LSTM-SHAP framework for interpretable control-model error diagnosis, with framework validity verified through SHAP-guided error correction.
• Designed a nonstandard five-degree-of-freedom welding robot based on the Pieper criterion and derived its closed-form inverse kinematics.
• Proposed a second-order differential feature-point extraction algorithm for laser-line grayscale images and optimized it using extensive experimental data.
• Developed a three-dimensional system-error model based on point-cloud registration by integrating ICP and RANSAC to improve estimation accuracy.
- Combined Structured-Light Sensing for Intelligent Pipeline Welding** *Sep. 2024–Present*
Technology Transfer Project, Beijing, China
• Trained a YOLO11 model for laser-line feature recognition and developed Weld-TRINet for real-time feature-point estimation.
• Conducted welding-data analysis and algorithm optimization, enabling automatic adjustment of torch pose and welding parameters in field operations.
• Co-designed a multimodal sensor integrating vision and force sensing and developed time-frequency-spatial image-processing algorithms.

PUBLICATIONS

- Zhu, C.; **Wang, Z.**; Zhu, Z. “Research on Pipeline Intelligent Welding Based on Combined Line Structured Lights Vision Sensing: A Partitioned Time-Frequency-Space Image Processing Algorithm.” *International Journal of Advanced Manufacturing Technology*, 2024, 134(11-12), 5463-5479. (Co-first author; SCI; JCR Q2.)
- Zhu, C.; **Wang, Z.**; Zhu, Z.; et al. “Research and Intelligent Development of Pipeline External Welding Equipment and Sensing Technology.” *Journal of Manufacturing Processes*, 2025, 152, 390-410. (SCI; JCR Q1; CAS Tier 1 Top Journal.)
- Zhu, C.; **Wang, Z.**; Zhu, Z.; et al. “Design and Modeling of Pipeline Welding Robot Based on Pieper Criterion and Construction of Integrated Intelligent Welding System.” *Welding in the World*, 2025, 69, 2853-2867. (SCI; JCR Q2.)
- Wang, Z.**; Zhu, C.; Zhu, Z.; et al. “Cross-Layer Trajectory Prediction and Fusion Control for Pipeline Welding Robots Based on Online Planning.” *Results in Engineering*, **under review**. (First Author; SCI, JCR Q1.)
- Wang, Z.**; Zhu, C.; Zhu, Z. “Classification and Parameter Optimization of Automatic Welding Tracking Errors Based on Explainable Temporal Learning.” (First author; manuscript in revision.)

PATENTS

- Zhu, Z.; **Wang, Z.**; Zhu, C. Numerical Method for Determining System Errors of Rail-Guided Welding Robots Based on Point-Cloud Registration. Chinese Invention Patent CN121837275B. Granted Jul. 28, 2026. (Student First Inventor)
- Zhu, Z.; Zhu, C.; **Wang, Z.** Online Planning Method and Related Equipment for Welding-Torch Pose in Multi-Layer Multi-Pass External Pipeline Welding. Chinese Invention Patent CN119387995B, granted Aug. 12, 2025.

TEACHING AND LEADERSHIP

- Teaching Assistant, Fundamentals of Control Engineering** *Sep. 2024–Jun. 2025*
Tsinghua University, Beijing, China
• Managed undergraduate attendance, graded assignments, answered course questions, and calculated final grades for the teaching team.
- Youth League Secretary and Academic Representative, Mechanical Engineering Class 02** *Sep. 2020 – Jun. 2023*
Tsinghua University, Beijing, China
• Helped the class maintain one of the highest academic standings in the cohort and achieve the cohort’s highest numbers of recommended master’s and direct-entry doctoral admissions.
• Contributed to the class receiving the Grade-A Youth League Branch distinction from the Department of Mechanical Engineering in 2020.

TECHNICAL SKILLS AND INTERESTS

- Programming: C++, Python, MATLAB
- Engineering Software: SolidWorks, AutoCAD
- Interests: Reading and essay writing